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Method for separating amoxicillin and phenylacetic acid from reaction solution in one-step enzymatic synthesis of amoxicillin

Abstract

A method for separating amoxicillin and phenylacetic acid from reaction solution in one-step enzymatic synthesis of amoxicillin is provided. The method employs immobilized penicillin acylase mutant to catalyze the one-step synthesis of amoxicillin from penicillin potassium, and develops a separation process for the resulting reaction mixture. The technical scheme mainly comprises: Firstly separating the immobilized penicillin acylase mutant from the reaction solution through filtration; subsequently isolating amoxicillin via crystallization; followed by separating and recovering phenylacetic acid through toluene extraction and back extraction. This separation method enables rapid and efficient isolation of amoxicillin with high production yield, achieving an average crystallization rate of 93.22%. Concurrently, it demonstrates effective separation and recovery of phenylacetic acid while allowing recyclable use of the toluene extractant.

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Background/Summary

CROSS REFERENCE TO THE RELATED APPLICATIONS

(1) This application is based upon and claims priority to Chinese Patent Application No. 202411522210.7, filed on Oct. 29, 2024, the entire of which are incorporated herein by reference.

SEQUENCE LISTING

(2) The instant application contains a Sequence Listing which has been submitted in XML format via EFS-Web and is hereby incorporated by reference in its entirety. Said XML copy is named FSJZ0102S_Sequence_Listing, created on 05/08/2025, and is 18,431 bytes in size.

TECHNICAL FIELD

(3) The present invention pertains to the technical field of product separation in the enzymatic synthesis of antibiotics, and more specifically relates to a method for separating amoxicillin and

phenylacetic acid from a reaction solution produced by a one-step enzymatic catalytic process for preparing amoxicillin.

BACKGROUND

(4) Amoxicillin (AMOX), also known as hydroxyampicillin, serves as a primary member of the second-generation penicillins. As a broad-spectrum semi-synthetic antibiotic, amoxicillin inhibits bacterial cell wall synthesis. Recognized by the World Health Organization (WHO) as the preferred β -lactam oral antibiotic due to its high efficacy, broad antimicrobial spectrum, and minimal adverse effects.

(5) Methods for amoxicillin preparation include chemical synthesis and enzymatic catalysis. Chemical synthesis suffers from prolonged reaction steps, high levels of three wastes (waste gas, wastewater, and solid residues), and excessive use of chemical solvents. With the advancement of green synthesis concepts in pharmaceutical manufacturing and improvements in enzymatic synthesis technology, enzymatic catalytic synthesis has become the predominant method for amoxicillin production. The primary process involves: synthesizing amoxicillin through penicillin acylase-catalyzed reaction of 6-APA with D-p-hydroxy-phenylglycine (or its methyl ester)(e.g., Chinese Patent Application No. 201711221286.6), followed by separation, purification, crystallization, and drying to obtain the final product.

(6) However, challenges persist in enzymatic amoxicillin synthesis. For instance, Chinese Patent CN102660621A discloses a process using 6-APA and D-p-hydroxy-phenylglycine methyl ester, but exhibits low final product yield and poor flowability. Penicillin acylase mutants enable one-step amoxicillin synthesis directly from penicillin and its salts. This technology eliminates multi-step reactions and intermediate product (e.g., 6-APA) separation processes required in conventional methods, offering simplified production, enhanced efficiency, and significantly reduced costs. A critical issue arises in the one-step synthesis reaction solution, which contains equimolar amounts of amoxicillin (AMOX) and phenylacetic acid (PAA). It is necessitating that high-yield isolation of amoxicillin meeting active pharmaceutical ingredient standards and efficient recovery and reuse of phenylacetic acid. Existing separation techniques for multi-step processes cannot meet the purification requirements for one-step synthesis.

SUMMARY

(7) To address the limitations of existing technologies, the present invention provides a production technology for efficiently and rapidly purifying amoxicillin and recovering phenylacetic acid, thereby achieving rapid separation and purification of amoxicillin and large-scale recovery of phenylacetic acid. The amoxicillin produced meets the requirements of pharmaceutical production and complies with active pharmaceutical ingredient standards.

(8) The present invention is implemented as follows: a method for separating amoxicillin and phenylacetic acid from a reaction solution obtained by preparing amoxicillin via one-step enzymatic catalysis. The method is characterized in comprising the steps of: (1) adding deionized water to the reaction suspension, and performing vacuum filtration to obtain an amoxicillin filtrate and a retained immobilized penicillin acylase mutant; (2) washing the retained immobilized penicillin acylase mutant with deionized water, and combining the wash solution with the amoxicillin filtrate from step (1); (3) adjusting the pH of the mixed solution obtained in step (2) to 2 with hydrochloric acid to obtain a separation-ready mixture; (4) adding a NaOH solution dropwise to the separation-ready mixture until reaching a pH between 3.5 and 5.5, followed by static crystallization at 4° C.; and (5) after crystallization is complete, filtering to obtain amoxicillin crystals and a liquid phase containing phenylacetic acid.

(9) Preferably, further comprising the following steps after step (5): (6) adjusting the pH of the liquid phase containing phenylacetic acid to between 2.0 and 2.5, and extracting with toluene to obtain an organic phase containing phenylacetic acid; (7) subjecting the organic phase containing phenylacetic acid to back-extraction with a NaOH solution, converting phenylacetic acid into sodium phenylacetate in an aqueous phase; (8) separating the aqueous phase from the organic

phase, wherein the toluene organic phase is recycled for extraction in step (6); (9) adding hydrochloric acid to the aqueous phase containing sodium phenylacetate to adjust the pH to between 2.0 and 2.5, converting sodium phenylacetate into phenylacetic acid, followed by crystallization at 4° C.; and (10) after crystallization is complete, filtering to separate and recover phenylacetic acid crystals.

(10) Preferably, in step (3), the hydrochloric acid concentration is 15% by volume.

(11) Preferably, in step (4), the pH for amoxicillin crystallization is 5, and the crystallization time is 9 hours.

(12) Preferably, in step (5), after filtration and separation, the amoxicillin crystals are dried in a vacuum oven at 50° C. for 2 hours.

(13) Preferably, in step (6), the pH is adjusted using 15% by volume hydrochloric acid; extraction with toluene is performed at least twice, and the resulting organic phases are combined.

(14) The method for catalyzing the one-step synthesis of amoxicillin from penicillin potassium using the immobilized penicillin acylase mutant comprises: using only one penicillin acylase mutant as a sole enzyme in a reaction system, using the penicillin or the salt thereof and D-p-hydroxyphenylglycine methyl ester as substrates, and carrying out a reaction in a buffer system of pH 4-8; wherein compared with the amino acid sequence shown in SEQ ID NO: 1, an amino acid sequence of the penicillin acylase mutant comprises at least one of the following mutations: F146αK, F24βR, F71βY, N241βK, G385βY, or G385βR.

(15) Process specifically includes the following steps: S1: adding a buffer solution of pH 4-8 to a reaction flask as a reaction buffer system; S2: adding a penicillin potassium salt and the D-p-hydroxyphenylglycine methyl ester to the reaction buffer system, wherein a molar ratio of the penicillin potassium salt to the D-p-hydroxyphenylglycine methyl ester is 1:1-1:2, and stirring thoroughly; and S3: adding an immobilized penicillin acylase mutant to the reaction buffer system, and reacting at a reaction temperature controlled at 12-30 C°. The preferred reaction temperature is 12 to 28° C.

(16) Preferably, wherein the buffer solution in the step S1 comprises any one of citric acid buffer, acetic acid buffer, phosphate buffered saline (PBS) buffer, sodium dihydrogen phosphate-citric acid buffer, sodium barbital-hydrochloric acid buffer, and pure water.

(17) Preferably, wherein in the step S2, a concentration of the penicillin potassium salt is 50-200 mmol/L, and a concentration of D-p-hydroxyphenylglycine methyl ester is 50-400 mmol/L.

(18) Preferably, wherein in the step S3, an amount of the immobilized penicillin acylase is 3000-30000 U/L, and a reaction time is 1-6 h.

(19) In summary, the advantages and positive effects of the present invention are as follows: the present invention uses an immobilized penicillin acylase mutant to catalyze the one-step synthesis of amoxicillin from potassium penicillin. A set of methods for separating amoxicillin and phenylacetic acid from the resulting reaction mixture has been developed. The technical scheme mainly includes the following: first, the immobilized penicillin acylase is separated from the reaction mixture by filtration; then, amoxicillin is isolated through crystallization; thereafter, phenylacetic acid is recovered through toluene extraction, back-extraction, and subsequent crystallization. This separation method enables rapid and efficient isolation of amoxicillin with a high yield-achieving an average crystallization rate of 93.22%. At the same time, the separation and recovery of phenylacetic acid is highly effective, and the extraction agent, toluene, can be reused.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic flow diagram illustrating a method for separating amoxicillin and phenylacetic acid from a reaction mixture obtained by one-step enzymatic synthesis of amoxicillin.

- (2) FIG. 2 shows the amino acid sequence (SEQ ID NO: 1) of the wild-type penicillin acylase enzyme.
- (3) FIG. 3 is a schematic diagram of the construction of the recombinant plasmid pET28a-kcPA.
- (4) FIG. 4 displays the agarose gel electrophoresis analysis of the PCR-amplified recombinant plasmid.
- (5) FIG. 5 presents the SDS-PAGE electrophoresis results of protein expression in *E. coli* BL21(DE3)/pET28a-kcPA cells.
- (6) FIG. 6 illustrates the HPLC chromatogram of amoxicillin synthesized in a one-step reaction catalyzed by KcPA from penicillin potassium salt in Embodiment 3.
- (7) FIG. 7 is a liquid chromatography chart of the amoxicillin crystal product.
- (8) FIG. 8 is a liquid chromatography chart of the phenylacetic acid crystal product.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (9) For clarity and to avoid limiting the scope of the invention, all numerical values expressing quantities, percentages, or other parameters in this application shall be interpreted as being modified by the term “about.” Unless explicitly stated otherwise, all numerical parameters in the specification and claims are approximations that may vary depending on the desired properties sought to be achieved. These numerical parameters should be interpreted in light of the reported significant digits and conventional rounding techniques. As used herein, “about” means within $\pm 10\%$ of a stated value or range, preferably within $\pm 5\%$.
- (10) Ambient temperature: Unless otherwise specified, experiments are conducted under ambient conditions (natural room temperature without additional cooling/heating), typically defined as 10 to 30° C., preferably 15 to 25° C. Abbreviations: “min” (minute), “s” (second), “U” (enzyme activity unit), “mM” (millimolar per liter), “M” (molar per liter), “rpm” (revolutions per minute), “mol” (mole), “ μg ” (microgram), “mg” (milligram), “g” (gram), “ μL ” (microliter), “mL” (milliliter), “bp” (base pair), “LB medium” (Luria-Bertani medium), “Kan50” (medium containing 50 $\mu\text{g/mL}$ kanamycin).
- (11) For experimental methods not specified with specific conditions in the examples, conventional conditions were typically followed, such as those described in the “Molecular Cloning: A Laboratory Manual” (Chinese Edition) (J. Sambrook, M. R. Green, translated by H. Fuchu, 4th edition, Beijing: Science Press, 2017) and in New England Biolabs (NEB) kits.
- (12) The present invention discloses a method for separating amoxicillin and phenylacetic acid from a reaction mixture obtained by one-step enzymatic synthesis of amoxicillin. The main content includes: first, using an immobilized penicillin acylase mutant to catalyze the one-step synthesis of amoxicillin from potassium penicillin, obtaining a reaction suspension, and then performing separation processing on the reaction suspension. The main steps include: (1) Under ambient temperature conditions, adding an equal volume of deionized water to the reaction suspension, filtering under normal pressure to obtain a filtrate containing amoxicillin and a retained immobilized penicillin acylase; (2) Washing the retained immobilized penicillin acylase with deionized water, combining the washing solution with the amoxicillin filtrate obtained in step (1); (3) Adjusting the pH of the mixed solution obtained in step (2) to 2 using hydrochloric acid, filtering to obtain an amoxicillin reaction solution; (4) Adjusting the pH of the amoxicillin reaction solution to 5.0 with a 0.25 M NaOH solution for crystallization, allowing the solution to crystallize at 4° C. for 9 hours. After completion of crystallization, separating solids from liquids to obtain amoxicillin crystals and a liquid phase containing phenylacetic acid; (5) Adjusting the pH of the liquid phase containing phenylacetic acid to between 2.0 and 2.5, extracting with toluene to transfer phenylacetic acid from the aqueous phase into the organic phase; (6) Adding a 0.25 M NaOH solution to the aforementioned organic phase to convert phenylacetic acid into sodium phenylacetate, which enters the aqueous phase; (7) Under heating conditions, adding 15% hydrochloric acid to the aforementioned aqueous phase containing sodium phenylacetate to adjust the pH to 2-2.5, converting sodium phenylacetate back into phenylacetic acid, and allowing it to

crystallize at 4° C. The process flowchart is shown in FIG. 1.

(13) The following will describe the technical solutions of the present invention clearly and completely in conjunction with the embodiments of the present invention.

Embodiment 1

(14) Construction of the Prokaryotic Expression for the Mutant of Penicillin Acylase from *Kluyvera citrophila*, and its Functional Characterization

(15) 1. Construction of Wild-Type PA Expression Vector pET28a-kcPA

(16) The wild-type penicillin acylase used in this Embodiment originates from *Kluyvera citrophila* ATCC 21285. Its amino acid sequence (SEQ ID NO: 1) comprises four domains, the sequence from the N-terminus to the C-terminus of the protein is as follows: Signal peptide: Positions 1-26. α -subunit: Positions 27-235 (209 amino acids). Linker peptide: Positions 236-289 (54 amino acids). β -subunit: Positions 290-846 (557 amino acids). Refer to FIG. 2 for domain annotations (single underline: α -subunit; wavy line: linker; double underline: β -subunit). The nucleotide sequence is provided as SEQ ID NO: 2.

(17) The recombinant plasmid pET28a-kcPA was constructed as illustrated in FIG. 3. Using genomic DNA from *K. citrophila* ATCC 21285 as a template, primers were designed based on the nucleotide sequence of PA (SEQ ID NO: 2): Forward primer (SEQ ID NO: 3): 5'-CGG/AATTCATGAAAAACCGCAATCGCAT-3'. Reverse primer (SEQ ID NO: 4): 5'-CCA/AGCTTTTAGCGCTGCACCTGCAGC-3'. EcoRI and HindIII restriction enzyme sites were introduced into the primer sequences (the underlined bases indicate the restriction enzyme recognition sites), and the target fragment of the wild-type PA was then amplified by PCR.

(18) TABLE-US-00001 PCR Reaction mixture Component Volume Sterile ddH₂O 10 μ L template 0.5 μ L Forward primer (10 μ M) 1 μ L Reverse primer (10 μ M) 1 μ L 2 $\times$ Taq Polymerase 12.5 μ L Total reaction volume 25 μ L

(19) TABLE-US-00002 PCR Thermal Cycling as following: {circle around (1)}. 95° C. 5 min {circle around (2)}. 95° C. 45 s {circle around (3)}. 60° C. 50 s {circle around (4)}. 72° C. 90 s {circle around (5)}. Go to {circle around (2)} 30 cycles {circle around (6)}. 72° C. 10 min {circle around (7)}. 4° C. forever

(20) The empty plasmid pET28a and PCR-amplified PA DNA fragment were subjected to double restriction enzyme digestion using EcoRI and HindIII restriction enzymes.

(21) TABLE-US-00003 Double enzyme digestion system: Component Volume/Amount 10x Buffer 5 μ L EcoRI 1.5 μ L HindIII 1.5 μ L Empty vector or 42 μ L PCR product Total 50 μ L

(22) The double restriction enzyme digestion was carried out at 37° C. for 1 hour, followed by enzyme inactivation at 80° C. for 20 minutes. The digested products were purified and quantified via agarose gel electrophoresis, yielding approximate concentrations of 50 ng/ μ L for the pET28a vector and 140 ng/ μ L for the kcPA insert.

(23) The purified fragments were ligated using T4 DNA ligase in a 16° C. metal bath overnight to generate the recombinant plasmid pET28a-kcPA. The recombinant plasmid was introduced into competent *E. coli* DH5 α cells via heat shock transformation.

(24) Connection Reaction System of the Target Fragment and the Linearized Vector:

(25) TABLE-US-00004 Component Volume 10x Buffer 1 μ L T4 DNA ligase 1 μ L Linearized vector 4 μ L Insert fragment 4 μ L Total 10 μ L

(26) To verify successful plasmid transformation, single colonies were picked from Kan50-LB agar plates and inoculated into Kan50-LB liquid medium. On the following day, plasmids were extracted using a plasmid extraction kit and subjected to PCR verification. Agarose gel electrophoresis confirmed the presence of a 2500-bp target band (see FIG. 4). The validated expression vector pET28a-kcPA was then transformed into *E. coli* BL21(DE3) to generate the recombinant strain *E. coli* BL21(DE3)/pET28a-kcPA for wild-type PA expression.

(27) 2. Generation of Mutant Expression Vectors

(28) In this embodiment, a total of 18 mutants were obtained through site-directed mutagenesis, as

shown in the table below. The notation “F146αK” indicates that the amino acid at position 146 on the α subunit was changed from F (Phenylalanine) to K (Lysine). The explanation for other mutation sites follows the same logic.

(29) TABLE-US-00005 TABLE 1 Mutants and Corresponding Mutation Sites
Mutant ID Mutation Sites
KcPA Wild-type KcPA 01-06 Single-point mutants: F146αK, F24βR, F71βY, N241βK, G385βY, G385βR
KcPA 07-11 Double-point mutants: F146αK combined with one of F24βR, F71βY, N241βK, G385βY, or G385βR
KcPA 12 Triple-point mutant: F146αK & F24βR & F71βY
KcPA 13 Triple-point mutant: F146αK & N241βK & G385βY
KcPA 14 Triple-point mutant: F146αK & N241βK & G385βR
KcPA 15 Triple-point mutant: F146αK & F71βY & N241βK
KcPA 16 Quadruple-point mutant: F146αK & F71βY & N241βK & G385βY
KcPA 17 Quintuple-point mutant: F146αK & F24βR & F71βY & N241βK & G385βY
KcPA 18 Quintuple-point mutant: F146αK & F24βR & F71βY & N241βK & G385βR

(30) First, primers corresponding to each mutation site were designed. Then, Using the wild-type PA target fragment as the initial template, site-directed mutagenesis was performed with the NEB Q5® Site-Directed Mutagenesis Kit (Q5 SDM Kit). The primers for each mutation site are listed below (lowercase letters indicate mutated nucleotides):

(31) TABLE-US-00006 F146αK, Forward (F): 5'-GGCGAACCGTaaaTCTGACAGCACCAG-3', SEQ ID NO: 5; Reverse (R): 5'-ATGGTGCCGACAAAAATCATCGCCA-3', SEQ ID NO: 6; F24βR, Forward (F): 5'-TGGGCGCGCAGcgcGGTTGGTATGCG-3', SEQ ID NO: 7; Reverse (R): 5'-TTGACCATAATGGCCTTCGCATCCT-3', SEQ ID NO: 8; F71βY, Forward (F): 5'-CACCGCCGGTtatGGTGTATGATG-3', SEQ ID NO: 9; Reverse (R): 5'-GATCCCCATGAAATGGTGCCGTTGT-3', SEQ ID NO: 10; N241βK, Forward (F): 5'-CGCCAACTGGaaaAACTCGCCGC-3', SEQ ID NO: 11; Reverse (R): 5'-ATATAGCCCGACTGCGGGTTATACAC-3', SEQ ID NO: 12; G385βY, Forward (F): 5'-CGGGCCAACctatTCGCTGAACATCAGCGTG-3', SEQ ID NO: 13; Reverse (R): 5'-TCCTGGGTGGTTTCATAGCCACTGG-3', SEQ ID NO: 14; G385βR, Forward (F): 5'-CGGGCCAACCcgcTCGCTGAACATC-3', SEQ ID NO: 15; Reverse (R): 5'-TCCTGGGTGGTTTCATAGCCACTGG-3', SEQ ID NO: 16;

(32) The primers were synthesized by a nucleic acid synthesis company, then dissolved in sterile water before proceeding with the protocol according to the kit instructions. As following:

(33) Step 1: Site-Directed Mutagenesis Via PCR

(34) TABLE-US-00007 PCR Reaction mixture: Component Volume
Q5 Hot Start High-Fidelity 2X 12.5 μL
Master Mix 10 μM Forward Primer 1.25 μL
10 μM Reverse Primer 1.25 μL
Template DNA (1-25 ng/μL) 1 μL
Nuclease-Free Water 9.0 μL
Total Volume 25 μL

(35) TABLE-US-00008 Thermal Cycling Program: Step Temperature Time
Initial 98° C. 30 s
Denaturation 98° C. 10 s
25 Cycles 68° C. 10-30 s 72° C. 62 s
Final Extension 72° C. 2 min
Hold 4-10° C. ∞

(36) For mutants with ≥2 mutation sites, the PCR product from the prior mutation step was used as the template for subsequent rounds of site-directed mutagenesis.

(37) Step 2: Kinase, Ligase & DpnI (KLD) Treatment

(38) TABLE-US-00009 Reaction mixture: Final Component Volume Concentration
PCR Product 1 μL 2x KLD Reaction Buffer 5 μL
1x 10x KLD Enzyme Mix 1 μL 1x Nuclease-Free Water 3 μL

(39) Incubate at room temperature for 5 minutes.

(40) Step 3: Heat Shock Transformation

(41) Add 5 μL of the KLD reaction mixture to 50 μL of chemically competent *E. coli* BL21(DE3) cells. Incubate on ice for 30 minutes, apply a 42° C. heat shock for 30 seconds, and return to ice for 5 minutes. Subsequently, add 950 μL of SOC sterile liquid medium and incubate at 37° C. with gentle shaking for 1 hour. Spread 40-100 μL of the bacterial suspension onto Kan50-LB agar plates and incubate overnight at 37° C. The resulting single colonies represent the mutant expression strains, designated as *E. coli* BL21(DE3)/pET28a-kcPA01~18.

(42) Step 4: Mutant Verification

(43) Inoculate the obtained mutant expression strains into 25 mL of LB liquid medium containing Kan50. Incubate at 37° C. overnight. Use a plasmid extraction kit to isolate the plasmid. Send the plasmid to a third-party biotech company for sequencing to confirm that the product is the intended point-mutated target product.

(44) 3. Expression of Wild-Type and Mutant KcPA

(45) The recombinant strains *E. coli* BL21(DE3)/pET28a-kcPA and *E. coli* BL21(DE3)/pET28a-kcPA01~18 were inoculated onto Kan50 LB agar plates and incubated in a 37° C. incubator for 12-16 hours. Single colonies were picked and inoculated into 25 mL of LB liquid medium containing Kan50. These cultures were grown overnight at 37° C. with shaking at 300 rpm. Next, 500 µL of the bacterial suspension was transferred to 50 mL of Kan50 LB liquid medium and incubated at 37° C. with shaking at 280 rpm. The OD600 was monitored, and when it reached 0.6-0.8, IPTG was added to a final concentration of 0.3 mM to induce protein expression. The cultures were induced for 10 hours at 25° C. with shaking at 220 rpm. The cells were harvested by centrifugation, and the cell pellets were resuspended in pre-chilled PBS buffer (pH 7.5) and kept on ice for 10 minutes. The resuspended cells were then centrifuged at 4° C., 12,000 rpm for 6 minutes to collect the cell pellets. The cell pellets were further resuspended in 50 mM PBS buffer (pH 7.5), and after another round of centrifugation at 4° C., 12,000 rpm for 6 minutes, the supernatant was discarded, and the final cell pellets were resuspended at a concentration of 0.01 g/mL. The cells were lysed using a sonicator under the following conditions: ice-water bath, 400 W power, with cycles of 3 seconds on and 5 seconds off for a total of 80 cycles. After lysis, the mixture was centrifuged at 4° C., 12,000 rpm for 15 minutes, and the supernatant was collected as the crude enzyme extract. SDS-PAGE was then used to analyze the expressed proteins.

(46) FIG. 5 shows the SDS-PAGE image of expressed proteins. The lanes are labeled as follows: M: Protein marker. Lane 1: Supernatant of *E. coli* BL21(DE3)/pET28a. Lane 2: Supernatant of *E. coli* BL21(DE3)/pET28a-kcPA without induction. Lane 3: Supernatant of *E. coli* BL21(DE3)/pET28a-kcPA18 without induction. Lane 4: Supernatant of IPTG-induced *E. coli* BL21(DE3)/pET28a-kcPA18.

Embodiment 2

(47) 1. Measurement of KcPGA Hydrolytic Activity

(48) Principle of Measurement: Penicillin Potassium Salt (PGK) is hydrolyzed under the action of KcPA to produce 6-Aminopenicillanic Acid (6-APA) and phenylacetic acid. Under acidic conditions, 6-APA reacts with p-Dimethylaminobenzaldehyde (PDAB) to form a yellow-green substance that has a maximum absorption peak at 415 nm. The enzyme activity is defined as: in 0.1 M PBS buffer at 28° C., the amount of penicillin acylase required to catalyze the conversion of 20 mg/mL PGK into 1 µmol of 6-APA per minute is defined as KcPA enzyme activity one unit (U).

(49) Weigh 0.5 g of PGK and dissolve it in the aforementioned buffer solution, then adjust the volume to 25 mL. Pipette 2 mL of PGK solution into a centrifuge tube and add 0.1 mL of KcPA enzyme solution. Set up a control without adding KcPA while keeping all other conditions identical.

(50) Place the reaction system in a water bath shaker at 28° C. and 200 rpm for 10 minutes. After the reaction, deactivate the enzyme by placing it in a 90° C. water bath for 2 minutes. Then, take 200 µL of the post-reaction solution and add it to 3 mL of 0.1 M citrate buffer at pH 3.0, followed by the addition of 1 mL of coloring reagent (0.5% PDAB). Allow it to stand at room temperature for 3 minutes before measuring the absorbance at 415 nm. Use the 6-APA standard curve to determine the concentration of 6-APA in the sample, and calculate the enzyme activity, i.e. the hydrolytic activity, using the formula provided.

(51) Calculation Formula: Hydrolytic Activity of Penicillin Acylase Per mL:

$$(52) U = \frac{C_{6-APA} \times V}{t \times V_E}$$

(53) Where, C.sub.6-APA: Concentration of 6-APA in the sample, µmol/L; V: Volume of the

reaction system, mL; V.sub.E: Volume of penicillin acylase added, mL; t: Reaction time, 10 min.

(54) 2. Measurement of KcPA Synthesis Activity

(55) 6-Aminopenicillanic Acid (6-APA) and D-p-Hydroxyphenylglycine methyl ester (D-HPGM) are catalyzed by KcPA to synthesize Amoxicillin. The amount of Amoxicillin can be measured using High-Performance Liquid Chromatography (HPLC), thereby calculating the synthesis activity of Penicillin Acylase (PA). The enzyme activity is defined as: under certain conditions, the amount of enzyme required to catalyze the production of 1 μ mol of Amoxicillin per minute is considered one unit of synthesis activity, denoted as U.

(56) Weigh 1 g of 6-APA and 1.25 g of D-HPGM and dissolve them in 50 mL of 0.1 M PBS buffer at pH 6.3. Adjust the pH to 6.3 and then make up to a total volume of 100 mL with the same buffer. Add 0.1 mL of KcPA enzyme solution to this mixture and incubate at 25° C. with shaking at 200 rpm for 30 minutes. Inactivate the enzyme by placing it in a 90° C. water bath for 2 minutes. Filter 0.5 mL of the reaction mixture through a 0.22 μ m aqueous filter membrane and dilute with phosphate buffer to 100 mL for HPLC analysis to determine the concentration of Amoxicillin. The formula for calculating enzyme activity is provided below:

(57) Calculation Formula: Synthesis Activity of Penicillin Acylase Per mL:

$$(58) U = \frac{C_{\text{sample}} \times V \times 200}{V_E \times t}$$

(59) Where, C.sub.sample: Concentration of Amoxicillin, μ mol/L; V: Volume of the reaction mixture, mL; 200: Dilution factor; V.sub.E: Volume of enzyme added, mL; t: Reaction time, min.

(60) HPLC Conditions: Agilent ZORBAX SB-C18 4.6 $\times$ 250 mm column, column temperature 25° C. Injection volume 10 μ L. Mobile phase A (0.02 M NaH.sub.2PO.sub.4—Na.sub.2HPO.sub.4 buffer at pH 4.7), mobile phase B (methanol). Start with 90% mobile phase A and 10% mobile phase B for 5 minutes, increase mobile phase B from 10% to 50% between 5 and 7 minutes, maintain 50% for 10 minutes, reduce mobile phase B from 50% to 10% between 17 and 19 minutes, and finally equilibrate with 90% mobile phase A and 10% mobile phase B for 5 minutes. Total flow rate is 1 mL/min.

(61) TABLE-US-00010 TABLE 2 Comparison of Activities Between Mutants and Wild Type
Comparison of hydrolysis Synthesis
Mutant ID activity Activity KcPA 100 100
KcPA 01 580 1530 KcPA 02 245 950 KcPA 03 286 838 KcPA 04 290 818 KcPA 05 492 709 KcPA 06 462 1118 KcPA 07 745 2054 KcPA 08 786 3037 KcPA 09 664 2260 KcPA 10 858 3574 KcPA 11 920 4380 KcPA 12 730 3410 KcPA 13 953 8842 KcPA 14 1074 10568 KcPA 15 620 4875 KcPA 16 831 8640 KcPA 17 1270 12680 KcPA 18 1440 15620

(62) Note: The hydrolytic activity of recombinantly expressed wild-type KcPA is 15 U/mL (fermentation broth), and the synthesis activity is 80 U/mL. For ease of comparison, the enzyme activity of the wild-type KcPA is defined as 100 in Table 2, with each mutant's activity compared against this baseline.

(63) From the table, it is evident that single-site mutants exhibit significantly enhanced hydrolytic and synthesis activities compared to the wild type, particularly F146 α K on the α subunit and G385 β R on the β subunit. Specifically, the single-point mutation F146 α K shows 5.8 times higher hydrolytic activity and 15.3 times higher synthesis activity than the wild type. The G385 β R mutant exhibits 4.6 times higher hydrolytic activity and approximately 11.2 times higher synthesis activity than the wild type. Comparing G385 β Y and G385 β R mutants, the former has higher hydrolytic activity but not superior synthesis activity. When multiple mutations are combined, the enzyme activity increases significantly over single mutations, especially for the five-point mutant F146 α K & F24 β R & F71 β Y & N241 β K & G385 β R, which shows high levels of both hydrolytic and synthesis activities.

Embodiment 3

(64) One-Step Synthesis of Amoxicillin Catalyzed by Wild-Type and Penicillin Acylase Mutants Using PGK

(65) In PBS buffer at pH 7.0, PGK was added to reach a concentration of 200 mM, along with D-p-

Hydroxyphenylglycine Methyl Ester (D-HPGM) to achieve a final concentration of 300 mM. Enzyme was added at 30 U/mL (based on synthetic activity). The reaction was carried out at 28° C. with constant stirring for 3 hours. After the reaction, HPLC analysis was performed to calculate the yield of Amoxicillin.

(66) HPLC Conditions: Agilent ZORBAX SB-C18 4.6×250 mm column, column temperature 25° C. Injection volume 10 µL. Mobile phase A (0.02 M NaH.sub.2PO.sub.4—Na.sub.2HPO.sub.4 buffer at pH 4.7), mobile phase B (methanol). Start with 90% mobile phase A and 10% mobile phase B for 5 minutes, increase mobile phase B from 10% to 50% between 5 and 7 minutes, maintain 50% for 10 minutes, reduce mobile phase B from 50% to 10% between 17 and 19 minutes, and finally equilibrate with 90% mobile phase A and 10% mobile phase B for 5 minutes. Flow rate is 1 mL/min.

(67) The reaction scheme is as follows:

(68) ##STR00001##

(69) For mutant KcPA 18, the HPLC detection spectrum is shown in FIG. 6, where D-HPG represents D-p-Hydroxyphenylglycine, AMOX represents Amoxicillin, D-HPGM represents D-p-Hydroxyphenylglycine Methyl Ester, PAA represents Phenylacetic Acid, and PGK represents Penicillin Potassium Salt. As can be seen from the figure, the content of the intermediate 6-APA is extremely low, almost negligible.

(70) TABLE-US-00011 TABLE 3 Yield of Amoxicillin Catalyzed by Various Mutants Amoxicillin Mutant ID Yield/% KcPA 8.5 KcPA 01 24.5 KcPA 02 16.3 KcPA 03 18.6 KcPA 04 17.2 KcPA 05 15.7 KcPA 06 20.1 KcPA 07 26.8 KcPA 08 28.5 KcPA 09 25.3 KcPA 10 29.2 KcPA 11 34.7 KcPA 12 28.9 KcPA 13 68.2 KcPA 14 85.6 KcPA 15 37.7 KcPA 16 67.9 KcPA 17 92.8 KcPA 18 99.0

(71) From the results in the table above, it is evident that all mutants are capable of catalyzing the reaction between potassium salt of penicillin and D-Hydroxyphenylglycine Methyl Ester in one step to synthesize Amoxicillin within a single reaction system. Moreover, the product yields are significantly higher compared to the wild type.

Embodiment 4

(72) Preparation of Immobilized Penicillin Acylase

(73) Penicillin acylase enzyme solution was prepared according to Embodiment 1. This enzyme solution was then added to epoxy resin (ER) activated with glutaraldehyde, and the cross-linking reaction was carried out at 15° C. for 1.5 hours. The cross-linking reaction system consisted of: 1200 U of enzyme solution, 0.25% glutaraldehyde, 5 g of ER, and 50 mL of phosphate buffer (KH.sub.2PO.sub.4—K.sub.2HPO.sub.4) at pH 7.5. After the completion of the cross-linking reaction, the immobilized enzyme was collected by filtration using a sieve. The immobilized enzyme was then washed with 100 mL of phosphate buffer (pH 7.5). The activity of the immobilized enzyme was measured to be 180 U/g, with an immobilized enzyme activity recovery rate reaching 75%.

Embodiment 5

(74) One Step Synthesis of Amoxicillin from PGK Catalyzed by KcPA 18

(75) (1) In a reaction flask, 50 mL of PBS buffer at pH 7 was added as the reaction buffer system. (2) A certain amount of Penicillin Potassium Salt (PGK) and D-p-Hydroxyphenylglycine Methyl Ester (D-HPGM) was accurately weighed and introduced into the reaction buffer system. The molar ratio of PGK to D-HPGM was in the range of 1:1 to 1:2. The concentration of PGK was in the range of 50-200 mmol/L, and the concentration of D-HPGM was in the range of 50-400 mmol/L. The mixture was then thoroughly stirred to ensure uniform dispersion of the substrates PGK and D-HPGM in the reaction system. In this example, the final concentrations of PGK and D-HPGM were 200 mmol/L and 300 mmol/L, respectively. (3) The penicillin acylase mutant enzyme solution, KcPA18, was immobilized according to the method described in Embodiments 4. A known amount of the immobilized enzyme was accurately weighed and added to the reaction flask. The enzyme dosage was 20 U/mL (calculated based on synthetic activity). The reaction

temperature was controlled at 24° C., and the reaction was carried out for 4 hours, yielding a milky-white reaction suspension.

Embodiment 6

(76) Separation and Recovery of Amoxicillin and Phenylacetic Acid

(77) 1. Separation and Crystallization of Amoxicillin

(78) (1) To 50 mL of the reaction suspension obtained in Example 5, an equal volume of deionized water was added. (2) The immobilized penicillin acylase was separated by filtration, followed by washing with 50 mL of deionized water. The filtrate and washings were collected and combined to a total volume of 150 mL. (3) 15% hydrochloric acid was added dropwise to the filtrate until the pH reached approximately 2, yielding a mixture ready for separation. (4) 0.25 mol/L NaOH solution was then added dropwise to the mixture until the pH reached 5. The mixture was placed in a refrigerator at 4° C. for crystallization over 9 hours. (5) After crystallization, the mixture was filtered, yielding amoxicillin crystals and a liquid phase containing phenylacetic acid. The amoxicillin crystals were dried in an oven at 85° C. for 30 minutes.

(79) To verify if the crystallized product is indeed amoxicillin, 0.1 g of the crystalline product was dissolved in deionized water and diluted to 50 mL, resulting in a 2 mg/mL solution. High-performance liquid chromatography (HPLC) analysis was performed under conditions identical to those described in Example 3. As shown in FIG. 7, no significant impurity peaks other than the AMOX peak were observed, confirming that the crystal is primarily amoxicillin.

(80) 2. Separation and Crystallization of Phenylacetic Acid

(81) (1) The liquid phase containing phenylacetic acid (150 mL) was adjusted to a pH between 2.0 and 2.5 by adding 15% hydrochloric acid. (2) 40 mL of toluene was added as an extraction solvent. Extraction was performed at 25° C. with stirring for 15 minutes followed by settling to allow layer separation, transferring phenylacetic acid from the aqueous phase to the organic phase (toluene). This extraction process was repeated twice, and the organic phases were combined. (3) In the collected organic phase, 30 mL of 0.25 mol/L NaOH solution was added and thoroughly mixed. The mixture was allowed to settle (or centrifuged) to separate the layers, converting phenylacetic acid into sodium phenylacetate which was transferred back to the aqueous phase. The aqueous phase was separated from the organic phase, with the latter being recycled for further extractions in step (2). (4) The aqueous phase from the previous step (lower phase) was adjusted to a pH between 2.0 and 2.5 using 15% hydrochloric acid while heating at 60° C. It was then transferred to 4° C. for crystallization over 12 hours, converting sodium phenylacetate back into phenylacetic acid and allowing it to crystallize. (5) After crystallization, the mixture was filtered, and the residue was phenylacetic acid crystals. These crystals were dried in a vacuum oven at 40° C. for 30 minutes.

(82) To verify if the crystallized product is indeed phenylacetic acid, 0.1 g of the crystalline product was dissolved in deionized water and diluted to 50 mL, resulting in a 2 mg/mL solution. HPLC analysis was performed under conditions identical to those described in Example 3. As shown in FIG. 8, no significant impurity peaks other than the PAA peak were observed, confirming that the crystal is primarily phenylacetic acid.

(83) The above examples are merely preferred embodiments of the present invention and should not be construed as limiting the scope of the invention. Any modifications, substitutions, and improvements made within the spirit and principles of the invention are intended to be included within the scope of protection of the invention.

Claims

1. A method for separating amoxicillin and phenylacetic acid from a reaction solution obtained by a one-step enzymatic synthesis of amoxicillin, comprising catalyzing a one-step synthesis of amoxicillin from penicillin potassium using an immobilized penicillin acylase mutant to obtain a reaction suspension; and subjecting the reaction suspension to a separation process comprising the

following steps: (1) adding deionized water to the reaction suspension, and performing vacuum filtration to obtain an amoxicillin filtrate and a retained immobilized penicillin acylase mutant; (2) washing the retained immobilized penicillin acylase mutant with deionized water, and combining a wash solution with the amoxicillin filtrate from step (1) to obtain a mixed solution; (3) adjusting a pH of the mixed solution obtained in step (2) to 2 with hydrochloric acid to obtain a separation-ready mixture; (4) adding a NaOH solution dropwise to the separation-ready mixture until reaching a pH between 3.5 and 5.5, followed by static crystallization at 4° C.; and (5) after the static crystallization is complete, filtering to obtain amoxicillin crystals and a liquid phase containing phenylacetic acid; wherein a method for catalyzing the one-step synthesis of amoxicillin from the penicillin potassium using the immobilized penicillin acylase mutant comprises: using only one immobilized penicillin acylase mutant as a sole enzyme in a reaction system, with penicillin or a salt thereof and D-p-hydroxyphenylglycine methyl ester as substrates, and carrying out a reaction in a reaction buffer system of pH 4-8; wherein compared with the amino acid sequence shown in SEQ ID NO: 1, the immobilized penicillin acylase has mutations: F146 α K and F24 β R and F71 β Y and N241 β Y and G385 β R.

2. The method according to claim 1, further comprising the following steps after step (5): (6) adjusting a pH of the liquid phase containing phenylacetic acid to between 2.0 and 2.5, and extracting with toluene to obtain an organic phase containing phenylacetic acid; (7) subjecting the organic phase containing phenylacetic acid to back-extraction with the NaOH solution, converting phenylacetic acid into sodium phenylacetate in an aqueous phase; (8) separating the aqueous phase from the organic phase, wherein the organic phase is recycled for extraction in step (6); (9) adding the hydrochloric acid to the aqueous phase containing sodium phenylacetate to adjust a pH to between 2.0 and 2.5, converting sodium phenylacetate into phenylacetic acid, followed by crystallization at 4° C.; and (10) after the crystallization is complete, filtering to separate and recover phenylacetic acid crystals.

3. The method according to claim 1, wherein in step (3), a hydrochloric acid concentration is 15% by volume.

4. The method according to claim 1, wherein in step (4), the pH for the static crystallization is 5, and a crystallization time is 9 hours.

5. The method according to claim 1, wherein in step (5), after filtration and separation, the amoxicillin crystals are dried in a vacuum oven at 50° C. for 2 hours.

6. The method according to claim 2, wherein in step (6), the pH is adjusted using 15% by volume hydrochloric acid.

7. The method according to claim 2, wherein in step (6), extraction with toluene is performed at least twice, and resulting organic phases are combined.
